

Technical Datasheet — VBF-T5R3-DSH-01

****Product**:** NMC811 / graphite flagship-vehicle pouch cell (virtual design, DFN-verified) · ****Date**:** 2026-08-26 · ****Case**:** t5_r3

Field	Value	Source
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Rated capacity (Ah)	5.0 Ah nominal (parameter set); 5.0442 Ah simulation-verified (1C, DFN)	`param_dump.txt`; `cell/r6_y4_1c_dfn.json:capacity_ah`
Nominal voltage / voltage window (V)	Window 2.5 – 4.2 V; 1C midpoint 3.832 V	`param_dump.txt`; `cell/r6_y4_energy_dfn.json:midpoint_voltage_v`
Rated energy (Wh)	18.2855 Wh (1C discharge integration)	`cell/r6_y4_energy_dfn.json:energy_wh`
Energy density	**535.72 Wh/kg** (contract caliber, electrolyte excluded); 995.79 Wh/L	`cell/r6_y4_energy_dfn.json:energy_density_wh_kg` / <code>energy_density_wh_l`</code>
Maximum continuous discharge rate	1C (5 A) verified by 1C_discharge protocol at 298.15 K; higher-rate discharge not simulated	`r6_y4_1c_dfn.json`; 5C/lowT protocols not part of task metrics
Fast-charge capability	**4C CC (20 A): 0 -> ~69% SOC (3.48 Ah) in 626 s** , then 4.2 V cut-off; T_max 327.60 K; anode potential never below 0 V (min +0.0200 V) -> **no plating signature** . CV tail beyond cut-off not simulated (current decays below 4C; plating risk monotonically decreases with current — domain reasoning, not a simulated claim). Unmodified baseline clamps at ~26 s (0.14 Ah): this design extends the CC window ~24x	`r6_y4_4c_dfn.json:time_s/voltage_v/anode_potential_v/T_max_K`
Operating temperature range	Verified points: 298.15 K 1C discharge; 318.15 K ambient 4C charge (cell peak 327.60 K). Full-range map N/A (lowT/highT protocols not part of task metrics)	protocol definitions in `bda` runner
Cycle life	**Not simulated (requires aging model)** — not provided, must not be inferred	—
Safety determination	4C fast charge: no plating (min anode potential +0.0200 V (PASS)) and T_max 327.60 K <= 333.15 K (PASS) . Overcharge/nail/crush: N/A (beyond pure-simulation boundary, requires physical experiment)	`r6_y4_4c_dfn.json`
Dimensions and mass	Stack 178.8 μm × 0.1027 m² electrode area (0.065 m × 1.58 m); 34.13 g (contract, electrolyte excluded) / 40.28 g incl. electrolyte estimate (1.2 g/cm³, annotated); enclosure not provided	`r6_y4_energy_dfn.json:thickness_m/area_m2/mass_kg` + `param_dump.txt`

This datasheet is a virtual-simulation document. Every value is mechanically derived from command outputs; no physical cell has been built. True-compute endorsement skipped (`real\_compute=false`, recorded in log.jsonl endorse entry).